MIL-D-10329A(MU)

10 June 1965
SUPERSEDING
MIL-B-10329(CmlC)
31 May 1950

MILITARY SPECIFICATION

DYE, p-NITROPHENYLAZO-BETA-NAPHTHYLAMINE (B-1)

1. SCOPE

1.1 This specification covers one grade of p-nitrophenylazo-beta-naphthylamine (B-1).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

FSC 6820

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[illegible]ies of specifications, standards, drawings, and publications r[illegible]ed by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Appearance. The dye, p-nitrophenylazo-beta-naphthylamine, shall be a brown powder with a slight bronze cast (see 4.6.1).

3.2 Physical and chemical requirements. The materials shall conform to the requirements of table 1 when tested as specified therein.

TABLE I. Physical and chemical requirements

| Property | Requirements | | Test method |
|---|---|---|---|
| | Minimum | Maximum | |
| Melting point, °C | 182.0 | | 4.6.2 |
| Ash, percent | | 0.5 | 4.6.3 |
| Volatile matter, percent | | 0.5 | 4.6.4 |
| Insoluble in nitrobenzene, percent | | 0.1 | 4.6.5 |
| Particle size, percent through No. 200 sieve | 98.0 | | 4.6.6 |

3.3 Preproduction. When specified in the contract or order, a preproduction sample shall be produced for examination and test. The preproduction sample shall be manufactured using the same materials, methods and equipment as will be used during regular production (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except



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as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.2 Special provisions.

4.2.1 Alternative inspection. The supplier may utilize any alternative inspection which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. A lot shall consist of the p-nitrophenylazo-beta-naphthylamine which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot. (See 6.3)

4.4 Sampling for test. Containers shall be taken at random from each lot as follows:

| Lot size | Number of containers |
|---|---|
| 1 | 1 |
| 2 - 275 | 2 |
| 276 - 545 | 3 |
| 546 - 900 | 4 |
| 901 - 1345 | 5 |
| 1346 - 1875 | 6 |
| 1876 - 2500 | 7 |

For lots exceeding 2500 containers, the number required shall be calcualted using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = number of containers
N = lot size

A 1/4 pound sample shall be removed from each container selected and placed in a clean, dry container labeled to identify the lot and the container from which it was taken.



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4.5 Inspection provisions. Each sample shall be tested as specified in 4.6. The acceptance number is zero.

4.6 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 Appearance. Examine the sample visually for compliance with 3.1.

4.6.2 Melting point. Grind a small portion of the sample into a fine uniform powder. Introduce a portion of the ground sample into the open end of melting point tube (Available as a commercial product or formed by sealing one end of a thin walled capillary tube whose approximate length is 9 centimeters and outside diameter is 1.5 to 2.0 millimeters (mm). Tap the sealed end of the tube gently on a solid surface to pack the sample loosely in the sealed end of the tube to a height of 4 to 6 mm. Attach the tube to a calibrated mercury thermometer, graduated in intervals of no more than 0.5°C in such a manner that the sample is adjacent to the midpoint of the thermometer bulb. Suspend the thermometer and melting point tube assembly in a circulating, clear, non-volatile oil bath which has been preheated to 170°C. Temperature shall be increased at a rate of 1°C per minute. The temperature at which the sample begins to melt shall be recorded. When necessary, a correction shall be applied for the emergent stem of the thermometer.

4.6.3 Ash. Weigh approximately 2.0000 grams (g) of sample into a tared porcelain dish or crucible and ignite. When decomposition and evolution of gas is complete, transfer the dish or crucible to a muffle furnace, at dull red heat, for 30 minutes. Cool in desiccator and weigh. Calculate percent ash as follows:

$$\text{Percent ash} = \frac{100\ (A-B)}{W}$$

where: A = weight of dish or crucible and ash.
B = tare weight of dish or crucible, and
W = weight of sample.

4.6.4 Volatile matter. Determine volatile matter in accordance with Method 4071 of Federal Test Method Standard No. 141 except that the test temperature shall be 115° ± 5° C.



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4.6.5 Insoluble in nitrobenzene. Shake approximately 1.0000 g of sample in 100 milliliters (ml) of redistilled nitrobenzene at 23° ∱ 2°C for 15 minutes. Filter the solution through a tared Gooch crucible with asbestos pad. Rinse the flask (used to shake solution) with three 5-ml portions of nitrobenzene and use the rinsings to wash the insolubles collected in the Gooch crucible. In same manner, wash and rinse with two 5-ml portions of 95-percent ethanol. The washings shall be done at 23° ∱ 2°C. Dry to constant weight at 115° ∱ 5°C. Calculate residue as percent insoluble in nitrobenzene as follows:

$$\text{Percent insoluble in nitrobenzene} = \frac{100(A-B)}{W}$$

where: A = weight of crucible and insoluble matter,
B = tare weight of crucible, and
W = weight of sample.

4.6.6 Particle size. Determine particle size in accordance with Method 4101 of Federal Test Method Standard No. 141 except that the sieve used shall be No. 200 sieve conforming to RR-S-366.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. The p-nitrophenylazo-beta-naphthylamine shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Classification rules or regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The p-nitrophenylazo-beta-naphthylamine, covered by this specification, is intended for use with protective equipment.



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6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether a preproduction sample is required.
    (1) Quantity required.
    (2) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
    (3) Name and address of test facility and shipping instructions when testing is performed by the Government.
    (4) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially unifrom.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding - off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM Designation:E29).

6,5 Manufacturing directive. Edgewood Arsenal has prepared Manufacturing Directive No. 149 containing a suggested method of manufacturing p-nitrophenylamo-beta-naphthylamine.

Custodian:

Army - MU

Preparing activity:

Army - MU (EA)

Project No. 6820-A009



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